

Kaduna.lk

A Unified Smart Platform for Roadside Assistance and Automotive Services in Sri Lanka



The Problem

Sri Lanka has **NO dedicated digital platform** for roadside assistance



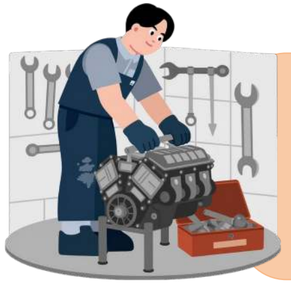
Stranded drivers have no help



No dispatch prioritization or congestion assessment

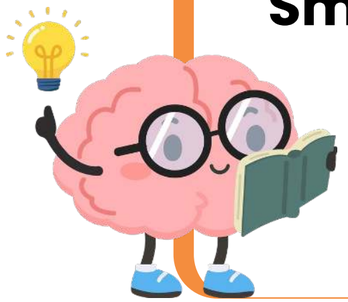


Mismatched providers drive out for nothing



No proper approach for the vehicle maintenance

Proposed Solution Overview



Smart Dispatch & Recovery

Works out what help you actually need — even when the problem is unknown

Geo-Intelligence & Traffic Impact



Scores how each incident affects traffic and prioritizes dispatch to ease congestion.

Predictive Maintenance



Reads live vehicle data to predict part wear and fix problems before they strand you.

Intelligent 3D Accident Claims



Rebuilds the accident scene in 3D from a phone — stopping recycled-photo claim fraud.

Stakeholders

Stranded Drivers — daily commuters, commercial operators



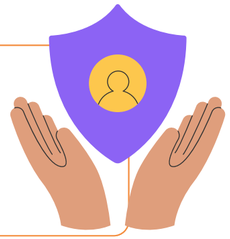
Roadside Assistance Providers — tow trucks, mechanics, fuel delivery



Insurance Companies — Licensed Insurance providers in Sri Lanka

Vehicle Service Centers — Registered repair centres

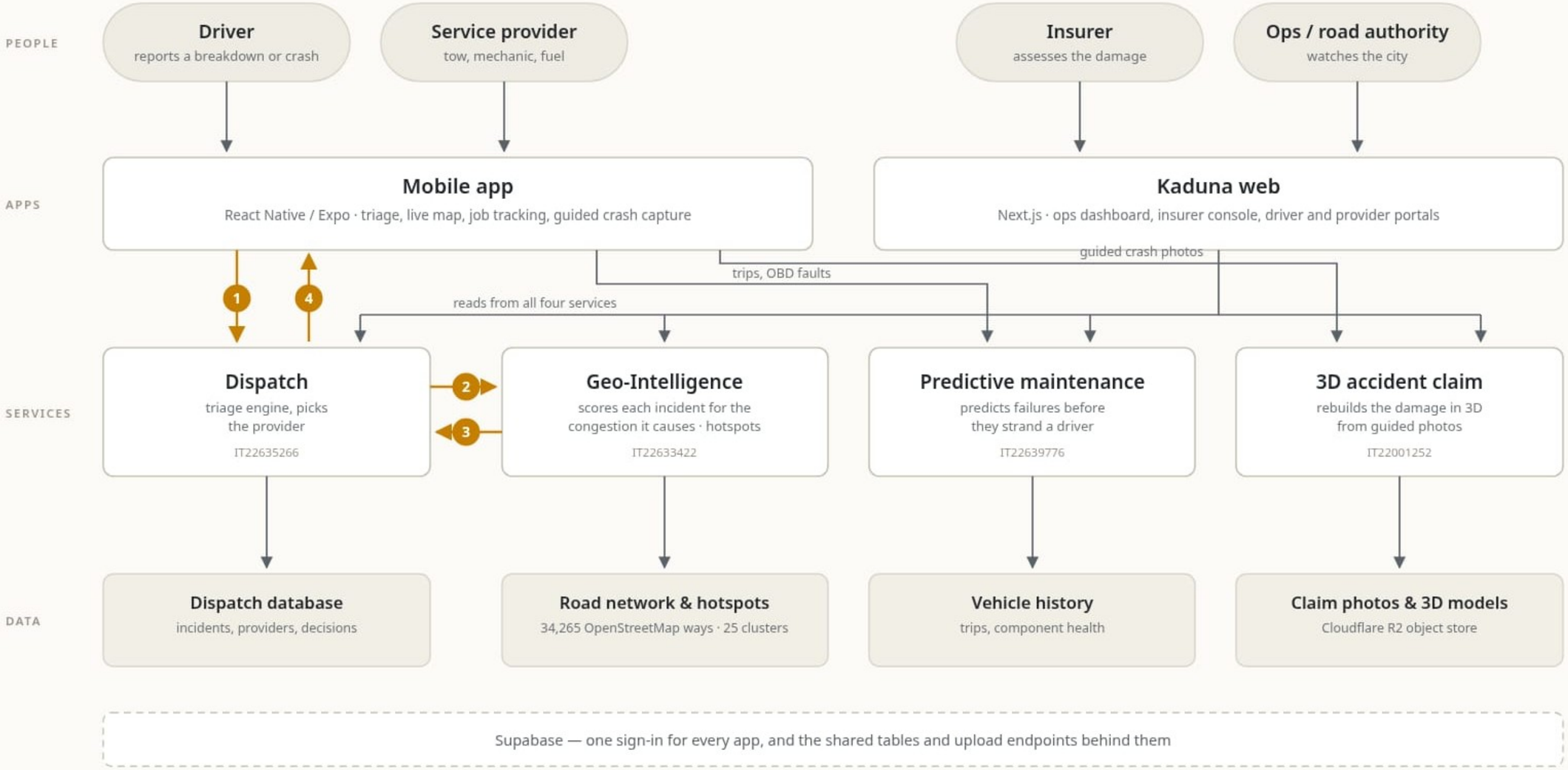
Traffic Authorities — Congestion monitoring and infrastructure planning



Research Gap & Objectives

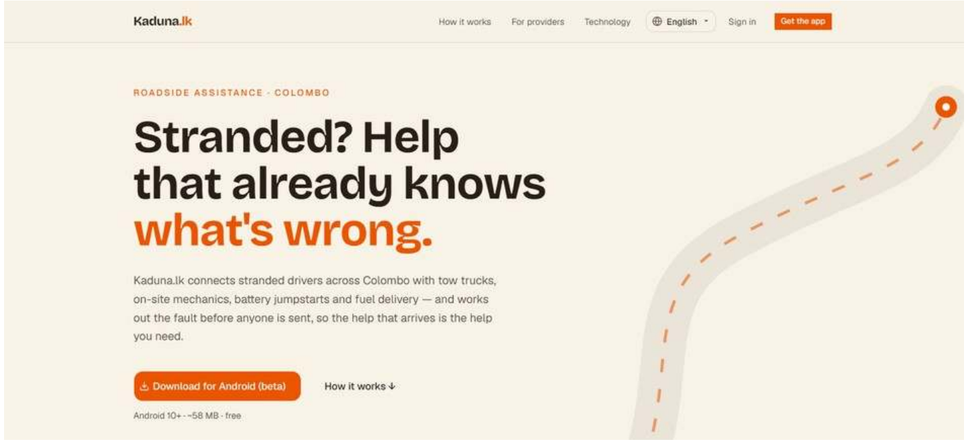
Component	Research Gap	Objective
Dispatch Optimization	No optimisation under diagnostic uncertainty, dispatch optimization, progressive learning	Stochastic dispatch over a distribution of service types, minimising expected cost, adaptive bayesian learning
Geo-Intelligence	No traffic impact model for Sri Lankan roads; dispatch ignores congestion Only Manual Hotspot analysis exist.	Multi-factor impact score (1–10) + Always recomputed current hotspot analysis + priority-based dispatch.
Predictive Maintenance	Isolated data sources; prediction without scheduling; built for modern fleets	Fuse telemetry + behaviour + service history → RUL for 4 components → cost-optimised schedules
3D Claims & Privacy	Photo-upload claims enable fraud, 2D night images unusable	3D/360° reconstruction + metadata verification, guidance capture, low light filters

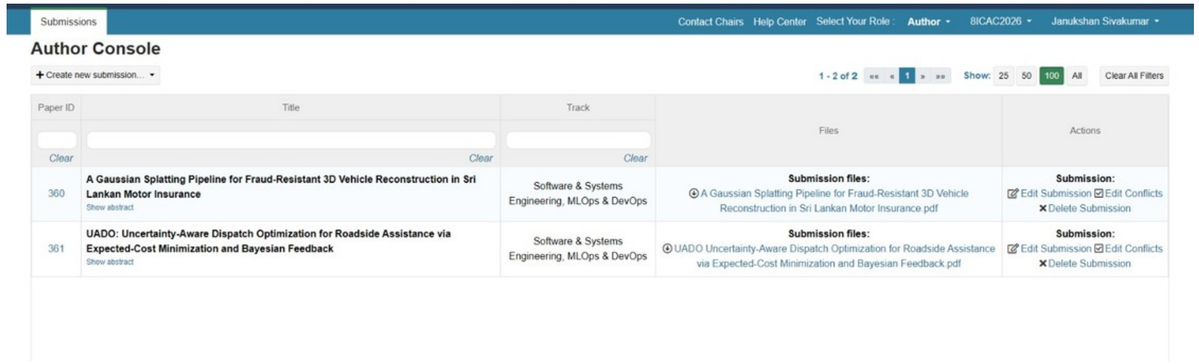
Architecture



Integration & Completion Status

- All four research components are fully integrated and work together.
- A unified mobile application has been developed for all components.
- All services are deployed and running in a production environment.
- Our official website and web application are now live.
- Ongoing collaboration with industry experts for validation and domain expertise.
- Two research papers have been registered for the ICAC Conference.





Commercialization

8M+
Registered Vehicles in Srilanka

1200 – 1500
Insurance Approved Garages

60 – 80
Service Center Franchise

*According to the **Sri Lanka Automobile Service Providers Association (SLASPA)** and the Department of Motor Traffic (DMT)*

Free Plan
Basic vehicle assistance features are available at no cost to drivers.

Premium Plan
Advanced features and benefits for power users and families.

B2B Subscription
Recurring subscription plans for fleet operators and insurance companies.

Service Booking Fees
Earn a commission on each booking made through the platform with garages and service centers.

Marketplace Commission
Earn a percentage from every product or service sold through the marketplace.

Advertising
Generate revenue from targeted promotions by parts suppliers and service brands.

Ongoing Collaborations

Automobile Association of Ceylon (AAC)

LAP 57 Boralegsgamuwa

Allianz Insurance Lanka

Department of Motor Traffic

Sanasa Insurance

Evaluation Evidence



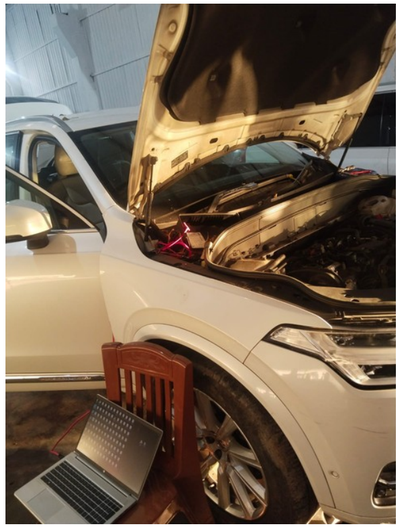
- Questionnaire and Fault scenarios validated by Lap 57.
- The Insurance claim functionality was validated by Sanasa Insurance.
- The app was given to several users and mechanics to use, and feedback was received.
- Traffic Impact Scoring and Prioritization guided by a retired Police DIG.
- OBD integration successfully tested in Toyota Aqua and a Audi A3.
- Fault codes and scenarios validated against Lap 57 Diagnosis History.

https://docs.google.com/spreadsheets/d/1oK3aqxQjBKDG33IjuYPTnZHog_IK4-H0HS9-3bFXpkg/edit?gid=0#gid=0









Non-Functional Requirements

- **Reliability/Availability** – Services are deployed on Contabo VPS infrastructure.
- **Security** – Supabase JWT-based authentication and Row-Level Security (RLS).
- **Scalability** – The PaaS-based deployment allows compute resources and service instances to be scaled as demand increases.
- **Maintainability** – The system follows a modular, service-based architecture, allowing individual research components/services to be updated, tested, and deployed independently.
- **Fault Tolerance** – Components are designed to degrade gracefully when dependent services are unavailable.
- **Usability** – A unified mobile application provides users with a guided workflow from incident reporting and diagnosis through service-provider dispatch.

Demo roadmap

1. Introduction to the Application
1. Fault Diagnosis and Service Provider Dispatching
1. Traffic Impact Scoring, Incident Hotspot Dashboard
1. Prioritize Hotspots and Locations
1. OBD Telemetry Integration, Predictive Maintenance
1. AI-Based Expert Support
1. 3D Model Demonstration, Lowlight Enhancement
1. How the Insurance Officers will use the system

Thank You !

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Questions & Discussion